Module 4: Intellectual Property Law & Data-Driven Innovation

This module explores the complex relationship between Intellectual Property (IP) law and the burgeoning data economy, focusing on whether EU law permits the mining of non-personal data from publicly accessible databases. The analysis covers the foundational principles of IP, the specific protections for databases, and the impact of recent legislation on Text and Data Mining (TDM).

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1. Foundational Concepts: IP Law and Data

a. What Can IP Law Protect? Intellectual Property law protects **products of the human intellect**. It does not protect raw, unorganized facts or data as such. Individual units of data (e.g., a single temperature reading, a number) are not considered products of human intellect and therefore cannot be protected by IP rights on their own.

However, when data is processed, organized, and structured to create meaningful **information**, it can, in certain situations, gain protection. - **Information as Trade Secrets:** Confidential business information can be protected under the EU Trade Secrets Directive (2016). - **Collections of Data as Databases:** Structured collections of data can be protected under the EU Database Directive (1996).

b. The Nature of Database Protection The **EU Database Directive (96/9/EC)** establishes a dual-protection (two-tier) system for databases, aiming to protect both creativity and investment. A single database can potentially qualify for both types of protection simultaneously.

A **database** is defined as "a collection of independent works, data or other materials arranged in a systematic or methodical way and individually accessible by electronic or other means."

- **Protection does not extend to the data itself:** The rights granted apply to the collection (the database), not the individual pieces of data contained within it.

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2. The Two Tiers of Database Protection

a. Tier 1: Copyright Protection for Original Databases This regime protects the creative effort invested in designing the database.

- **Subject-Matter:** The structure of the database.
- **Criterion for Protection: Originality.** The database must be the **"Author's Own Intellectual Creation" (AOIC)**. This standard is

met if the **selection or arrangement** of the database's contents reflects the author's free and creative choices, stamping the work with their "personal touch."
    - **Exclusion:** Protection is not granted if the structure is dictated by technical considerations, rules, or constraints that leave no room for creative freedom. Mere skill, labor, and know-how are insufficient.
- **Scope of Protection:** The copyright owner has exclusive rights over acts concerning the database's structure, including:
    - **Reproduction** (copying).
    - **Adaptation** (modification).
    - **Distribution** (sale).
    - **Communication to the public**.
- **Duration:** Lifetime of the author plus 70 years.
- **Practical Limitation:** Many modern digital databases, which are often comprehensive rather than selective and arranged logically (e.g., alphabetically) for functionality, fail to meet the AOIC originality standard and thus do not receive copyright protection.

b. Tier 2: The *Sui Generis* Database Right This unique right was created to protect the financial and labor investment in creating a database, even if it lacks originality.

- **Subject-Matter:** The contents of the database.
- **Criterion for Protection: Substantial Investment.** The maker of the database must demonstrate a "substantial investment" (financial, human effort, time, etc.) in either the **obtaining, verification, or presentation** of the contents.
    - **Crucial Distinction (The "Spin-off" Doctrine):** The investment must be for the collection of *existing* materials, not the *creation* of the data itself. If a database is a mere by-product ("spin-off") of a company's main activity that creates the data (e.g., a horse racing authority creating race data), the investment in creating that data does not qualify for *sui generis* protection. This significantly narrows the right's applicability.
- **Scope of Protection:** The database maker can prevent the:
    - **Extraction:** Transferring a substantial part of the contents to another medium.
    - **Re-utilisation:** Making a substantial part of the contents available to the public.
    - This also prohibits the **repeated and systematic extraction** of insubstantial parts if it conflicts with the normal exploitation of the database or unreasonably prejudices the maker's legitimate interests.
- **Duration:** 15 years. However, any new substantial investment in the database (e.g., a significant update) resets the 15-year clock, potentially leading to perpetual protection for dynamically updated databases.

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3. Text and Data Mining (TDM) and the Copyright in the Digital Single Market (DSM) Directive

The rise of AI and big data has made Text and Data Mining (TDM)—the automated analysis of large datasets to uncover patterns and trends—a critical tool for innovation. However, TDM inherently conflicts with database rights, as it often requires copying (reproduction) and accessing large amounts of content (extraction).

The **Directive on Copyright in the Digital Single Market (2019/790)** introduced two mandatory exceptions to address this conflict.

a. Article 3: TDM Exception for Scientific Research

- **Beneficiaries:** Limited to "research organisations" (like universities) and "cultural heritage institutions" (like public libraries and museums).
- **Scope:** Allows reproductions and extractions of lawfully accessed works for the purpose of scientific research.
- **Key Feature:** This exception is **unwaivable**. Rightholders cannot use contracts to override it.
- **Limitation:** It is of limited use for commercial entities like start-ups and SMEs, as they do not qualify as beneficiaries unless they are part of a public-private partnership, which can be impractical.

b. Article 4: General TDM Exception (The "Commercial" Exception)

- **Beneficiaries:** Open to everyone, including commercial entities, start-ups, and SMEs.
- **Scope:** Allows reproductions and extractions for any TDM purpose, including the development of commercial AI applications.
- **Major Weakness: The Rightholder's "Opt-Out".** The exception only applies if the rightholder has **not expressly reserved their rights**. Rightholders can easily opt out, for example, by using machine-readable means like a `robots.txt` file on a website or by stating it in their terms and conditions.
- **Impact:** This opt-out effectively creates a licensing market for TDM. It leaves commercial innovators in the EU at a competitive disadvantage compared to jurisdictions like the US, where TDM is often considered "fair use." Rightholders can prohibit TDM or charge for it, undermining the exception's goal of fostering innovation.

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4. Contract Law as a Barrier to Data Re-use

Beyond IP rights, contracts are a primary tool for controlling data access and use.

a. The *Ryanair v. PR Aviation* Case This landmark CJEU case addressed whether a database owner could contractually forbid the use of a database that was **not protected by either copyright or the *sui generis* right**. Ryanair's website terms and conditions prohibited "screen scraping" for commercial purposes. The court ruled that the Database Directive **does not prevent** a data holder from imposing contractual limitations on the use of an unprotected database.

b. Implications of the *Ryanair* Ruling

- **"Protection by the Back Door":** This decision allows data holders to use contracts to "retrieve" databases from the public domain, granting them a level of control that IP law explicitly denies. It creates a form of "private legislation."
- **Legal Uncertainty:** The enforceability of such contractual terms varies across Member States, undermining the harmony of the Digital Single Market.
- **Barrier to Innovation:** It reinforces the power of data holders to restrict data re-use, creating significant obstacles for start-ups and SMEs that rely on publicly available data for developing AI and other data-driven services. The new TDM exceptions in the DSM Directive do not solve this problem, as they only apply to databases protected by IP rights, not unprotected ones governed solely by contract.

5. Conclusion and The Way Forward

The current EU framework presents significant obstacles to data-driven innovation. While the Database Directive aimed to stimulate investment, its *sui generis* right has been criticized for creating data "lock-ins" without clear evidence of promoting database production. The new TDM exceptions in the DSM Directive are a step forward, but the commercial exception (Art. 4) is severely weakened by the opt-out provision.

Furthermore, the *Ryanair* ruling allows contracts to create stronger protection for unprotected databases than IP law provides for protected ones. This creates an imbalanced system that favors data holders over data users.

Potential reforms to create a more balanced framework include: 1. **Removing the Opt-Out from Article 4:** Making the commercial TDM exception mandatory and unwaivable by contract, at least for the *sui generis* right. 2. **Reforming or Abolishing the *Sui Generis* Right:** Given its questionable effectiveness and potential to hinder innovation, a radical reform (e.g., requiring registration) or complete abolition could be considered. 3. **Introducing a**

Data User Right: Creating a new, unwaivable right for users to re-use non-personal data from publicly disclosed databases that are not protected by IP. This would effectively reverse the *Ryanair* decision and ensure that data in the public domain remains freely re-usable.